

HackerRank Algorithmic Problem-Solving & Portfolio Integration

Name: Preetham Gowda G

Semester: 3rd Semester

Portfolio Links

GitHub Repository: <https://github.com/preethamgowda0702-spec/HackerRank-3rdSem-Portfolio>

HackerRank Profile: <https://www.hackerrank.com/profile/preethamgowda071>

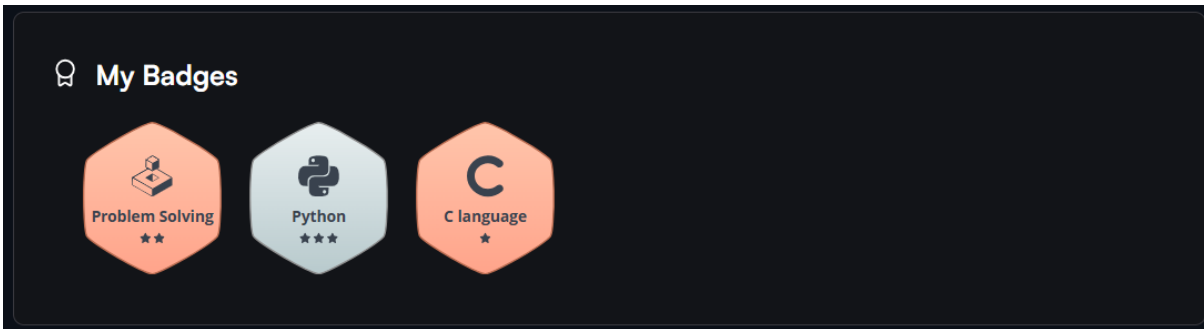
HackerRank Badge Information

C++: 2-Star Badge

Problems Completed

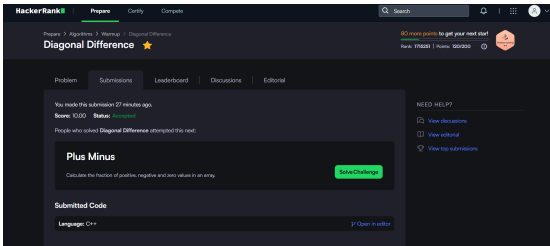
No.	Problem	Topic	Target Complexity
1	Diagonal Difference	2D Arrays	$O(N)$ / $O(1)$
2	Dynamic Array	Vectors	$O(N + Q)$ / $O(N)$
3	Time Conversion	Strings / Logic	$O(1)$ / $O(1)$
4	Compare the Triplets	Implementation	$O(1)$ / $O(1)$
5	Sparse Arrays	Hash Maps / Strings	$O(N + Q)$ / $O(N)$

HackerRank Profile / Badge Evidence

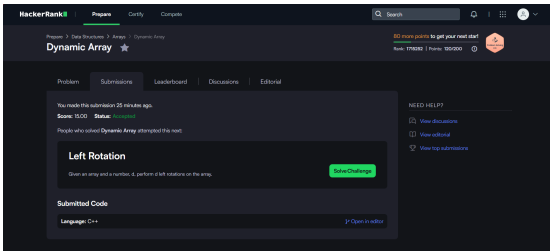


HackerRank profile badge section screenshot.

Accepted Submission Evidence

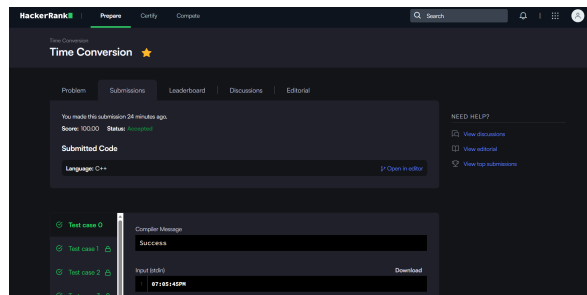


Diagonal Difference - Accepted

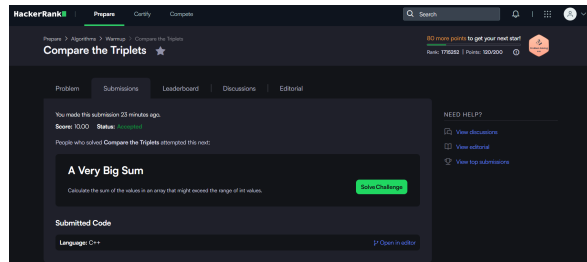


Dynamic Array - Accepted

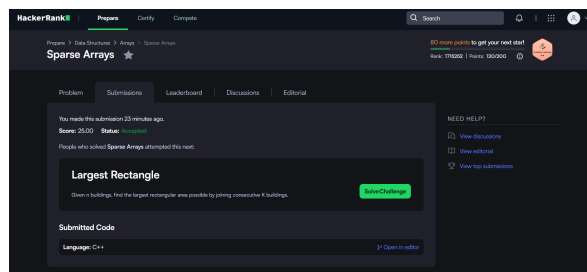
Accepted Submission Evidence



Time Conversion - Accepted



Compare the Triplets - Accepted



Sparse Arrays - Accepted

Reflection on Algorithmic Optimization Techniques

Through this HackerRank activity, I improved my understanding of algorithmic problem solving and efficient coding in C++. I solved five problems covering arrays, vectors, strings, matrix traversal, and hash maps. I learned that selecting an appropriate algorithm and data structure can reduce unnecessary operations and make programs more efficient.

While solving **Diagonal Difference**, I practiced matrix traversal and calculating diagonal sums efficiently. **Dynamic Array** helped me understand vectors, nested sequences, and bitwise XOR. **Time Conversion** improved my understanding of string manipulation and conditional logic. **Compare the Triplets** strengthened my comparison and implementation skills. **Sparse Arrays** introduced me to using hash maps for storing and retrieving frequencies efficiently.

I also learned to consider **time and space complexity** while writing solutions. Instead of focusing only on getting the correct output, I started thinking about how the solution performs as the input size increases. This activity improved my logical thinking, debugging skills, and familiarity with competitive programming. Uploading the solutions to GitHub also helped me organize my work and build a technical portfolio that demonstrates my programming and problem-solving practice.